

Sukhwinder Singh

B.E. Computer Science & Engineering — Thapar Institute of Engineering & Technology, Patiala
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Summary

Computer Science undergraduate (CGPA 8.41) with 5+ full-stack projects shipped using Spring Boot and React.js. Applied deep learning skills to event-camera depth estimation via a hybrid CNN–Transformer architecture. Solved 400+ DSA problems; seeking Software Engineering / GenAI internship roles.

Education

Year	Degree/Certificate	Institute	CGPA/%
2023–2027	B.E. Computer Science & Engineering	Thapar Institute of Engineering & Technology, Patiala	8.41 CGPA
2021–2023	CBSE (XII) — PCM	GMSSS, Sector 16, Chandigarh	91%
2019–2021	CBSE (X)	Manav Mangal High School, Chandigarh (Sector 21)	92.6%

Projects

CloudShare — Full-Stack Cloud Storage Platform

[GitHub](#) — [Live](#) — [Swagger](#)

Spring Boot, Spring Cloud Gateway, PostgreSQL, Redis, Cloudinary, React.js

- Architected 4 independent microservices (auth, file management, gateway, storage) with JWT authentication. Centralized routing through Spring Cloud Gateway.
- Configured Redis caching and Cloudinary storage to reduce repeat file-lookup latency across 12+ Swagger-documented endpoints.

CampusRide — Campus Cab Booking Platform

[GitHub](#) — [Live](#) — [Swagger](#)

Spring Boot, PostgreSQL, Spring Data JPA, React.js, Redis

- Delivered a cab booking system linking students and drivers under admin oversight. Secured endpoints with Spring Security and role-based JWT tokens.
- Optimized DTO-driven REST APIs with structured exception handling. Achieved a 120ms average response time across 30 endpoints in local load testing.

SkillBridge — Peer Skill-Barter Platform

[GitHub](#) — [Live](#) — [Swagger](#)

Spring Boot, WebSocket API, MongoDB, React.js

- Launched a barter-based exchange enabling users to trade skills directly. Built backend services for authentication, profile management, and skill discovery.
- Enabled real-time one-to-one chat via WebSocket and an admin moderation panel. Cut response-lookup time versus polling-based messaging.

EventSphere — Event Management & Ticketing Platform

[GitHub](#) — [Live](#) — [Swagger](#)

Spring Boot, PostgreSQL, Spring Data JPA, React.js, Redis

- Shipped an event and ticketing system with role-based access for organizers and attendees, supporting end-to-end booking workflows.
- Automated payment validation and email notifications across the booking workflow. Reduced manual confirmation steps to zero.

CodeE2LFDDepth — Hybrid CNN-Transformer Depth Estimation

[GitHub](#)

PyTorch, CNN, Swin-Transformer, Event-Camera Data

- Engineered a hybrid CNN–Transformer model with windowed (Swin-style) self-attention for event-camera light-field depth reconstruction. Added an A/B-testable variant on top of a baseline CNN pipeline.
- Automated PSNR/SSIM benchmark logging across 50 test scenes. Quantified the transformer variant’s gain over the CNN baseline.

Technical Skills

- **Languages:** C++, JavaScript, Java, Python, SQL
- **Frontend:** React.js, HTML5, CSS3, Bootstrap, JSX
- **Backend:** Node.js, Express.js, MongoDB, Spring Boot, Spring MVC, Hibernate, JDBC, Servlet, JSP
- **Databases:** MongoDB, MySQL, Firebase
- **ML/AI:** PyTorch, CNN/Transformer architectures, GenAI-assisted development workflows
- **Tools:** Git, GitHub, Maven, Postman, Vercel, Render, AWS, Figma, Canva, VS Code, IntelliJ IDEA

Positions of Responsibility

- **Head Boy, Manav Mangal High School** – Elected senior student representative. Led 10+ school-wide events as primary liaison between students and faculty.
- **Group Representative (3P1), TIET** – Coordinated schedules and resolved academic concerns. Served as liaison between students and instructors for a batch of [VERIFY: N] students.

DSA & Coding Profile

- Solved 400+ Data Structures & Algorithms problems across LeetCode, Code360, and GeeksforGeeks.
- Strong command over Arrays, Trees, Graphs, Dynamic Programming, Recursion, and Greedy algorithms. Proficient in Binary Search, Two Pointers, Sliding Window, and Hashing.

Relevant Coursework

Data Structures & Algorithms, Object-Oriented Programming, Computer Networks, Database Management Systems, Machine Learning (exploring)